AMARNATH S. PATEL

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Research interests: optical instrumentation & photonics, scientific/GPU computing, and machine learning for the physical sciences

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EDUCATION

University of Central Florida — **Undergraduate Student** — **4.00 GPA**

B.S. in Physics (Optics & Lasers) and Mathematics, Computer Science Minor (38 credit hours) — August 2025 – Present

Florida Atlantic University — **3.66 GPA**

University coursework completed via FAU High School, ages 14–18 (111 credit hours) — August 2021 – May 2025

RESEARCH EXPERIENCE

UCF Astrophotonics Lab, CREOL — Undergraduate Researcher — August 2025 – Present

Supervisor: Dr. Stephen Eikenberry

- Developing instrumentation and automation software for astrophotonics research spanning CREOL and the UCF Physics Department.
- Built an unattended pipeline that measures the wavelength-dependent complex transfer matrix of a photonic lantern via off-axis digital holography, coordinating 4 instruments across all-port × C-band (1525–1575 nm) sweeps at 92% reconstruction fidelity (building on the method of Dobias et al., *Opt. Express* 2026).
- Implemented the reconstruction chain: FFT sideband isolation and demodulation, Butterworth filtering, and numerically optimized quadratic-phase correction, feeding LP-mode decomposition to recover complex modal amplitude and phase.
- Developing PolyOculus, control and automation software for an 8-telescope photometric observation array; implementing INDI-protocol mount control, focuser automation, RA/Dec coordinate slewing, and backlash compensation.
- Developed Python SDK wrapper (gentec-camera) for an IR beam profiling camera enabling automated acquisition, real-time beam analysis, and FITS output.

UCF Physics Department — Paid Undergraduate Research Assistant — March 2026 – Present

Physics Education Research — Supervisor: Dr. Zhongzhou Chen

- Developing ESTELA, an automated system that generates multi-version isomorphic physics exams from AI-assisted problem banks for introductory STEM courses.
- Built in Rust with a vanilla JS frontend; parses YAML problem banks across 13 topic areas supporting 11 question types, renders LaTeX math via KaTeX, and exports multi-version exams as LaTeX or print-ready HTML/PDF.
- Funded by NSF award 2421299 and Gates Foundation INV-076932.

FAU Grant-Funded AI Safety Research Project (High School) — January 2024 – March 2025

Supervisor: Tucker Hindle, Florida Atlantic University

- Benchmarked 10+ coherence and detection methods (BERT NSP, GPT-2 perplexity, RoBERTa, LSA, NLI, burstiness) across 3,125+ generated sequences to evaluate approaches for identifying AI-generated text.
- Identified GPT-2 perplexity as the strongest discriminator (3.3× gap between coherent and incoherent text); BERT NSP failed to distinguish the two.
- Characterized a fundamental quality–evasion tradeoff across all humanization approaches tested, including iterative sampling, content abstraction, and RL-based methods.
- Grant-funded with HPC access; findings presented at the Wilkes Honors College Symposium (2025).

OTHER EXPERIENCE

Teaching Assistant — Calculus (High School) — August 2024 – May 2025

- Assisted 70 undergraduate students with learning calculus; office hours, exam review, grading. Part-time (10 h/week).

Advanced Experimental Vehicles — Programmer, Leader, Builder (High School) November 2023 – May 2025

- Configured Arch Linux ARM on Raspberry Pi 5 with Hyprland compositor and WireGuard VPN, enabling worldwide real-time telemetry monitoring of BMS, GPS, and camera feeds.
- Won 2nd Place in Division and Lockheed Martin Award for "Highest Level of Engineering Excellence."

PRESENTATIONS

Batch Processing for Automated Grading via Azure OpenAI June 2026

University of Central Florida, Downtown Campus. ESTELA project (Physics Education Research, Dr. Zhongzhou Chen).

Coherence and Detection Approaches for Identifying AI-Generated Text March 2025

Wilkes Honors College Undergraduate Research Symposium, Florida Atlantic University.

TECHNICAL REPORTS

Coherence and Detection Approaches for Identifying AI-Generated Text 2025

- Empirical study of AI-text detection and adversarial evasion: benchmarked 10 coherence and detection methods (GPT-2 perplexity best, 3.35× separation; RoBERTa the only effective detector) and built four evasion pipelines, characterizing a consistent quality–evasion tradeoff. Grant-funded, Florida Atlantic University.

SELECTED PROJECTS

CELERIS — GPU-Accelerated Metalens Design & EM Solver 2026

- From-scratch, validated electromagnetic solver for metalens design via rigorous coupled-wave analysis (RCWA): 1D TE/TM and 2D vectorial formulations, stable scattering-matrix recursion, and dispersion models (Sellmeier, tabulated n,k).
- CUDA-accelerated far-field analysis (600–700× over a 16-core CPU); inverse-design optimizer and fabrication-ready GDSII export, from a CLI or native desktop GUI. C++/CUDA with Python bindings.

coat — Cross-Platform Color-Scheme Configurator 2026

- Rust CLI that applies Base16/Base24 color schemes across 22 Linux applications and themes Windows system colors from a single config file; compatible with the 700-scheme tinted-theming ecosystem, with a terminal scheme browser and live RGB previews.

TECHNICAL SKILLS

Programming Languages: C/C++, CUDA, Rust, Python, JavaScript, Shell (Fish, Bash, tcsh), LaTeX/Typst

Scientific Computing: NumPy, SciPy, GPU/CUDA acceleration, FFT & signal processing, electromagnetic simulation (RCWA)

Tools & Libraries: Qt/PySide6, Docker, Git, CMake, XMake, INDI, KaTeX

Instrumentation: GPIB/VISA, RS-232, GigE Vision camera acquisition, tunable IR laser control, motorized polarization control, FITS & GDSII data formats

Operating Systems: Linux (Arch, NixOS, Ubuntu), BSD, Windows

RELEVANT COURSEWORK

University of Central Florida

Physics

Geometric Optics & Lab

Modern Physics [PHY 3101]

Quantum Information Processing
Independent Research [PHY 4912] (In Progress)
Mathematical Methods for Physics [PHZ 3113]
Electricity & Magnetism I (In Progress)

Mathematics

Applied Linear Algebra [MAS 3105]
Linear Algebra (Proof-Based) [MAS 3106] (In Progress)
Complex Analysis [MAA 4402]
Partial Differential Equations (In Progress)

Computer Science

C Programming [EGN 3211]
Object-Oriented Programming (In Progress)
Discrete Structures

Florida Atlantic University

Physics

General Physics I
General Physics II — Honors

Mathematics

Calculus I
Honors Calculus III
Elementary Matrix Algebra [MAS 2103]
Calculus II
Ordinary Differential Equations [MAP 2302]

Computer Science

Data Structures & Algorithms
Computer Architecture
C++ Programming [COP 3014]
Computer Logic Design
Deep Learning [CAP 4613]
Web Development [COP 3813]

HONORS & AWARDS

Lockheed Martin Award — "Highest Level of Engineering Excellence," AEV Competition 2024

2nd Place in Division, Advanced Experimental Vehicles Competition 2024

Florida Bright Futures — Florida Academic Scholars (highest tier; 100% tuition) 2025

NSF Award 2421299 — Research Funding (ESTELA project, with Dr. Zhongzhou Chen) 2026

Gates Foundation INV-076932 — Research Funding (ESTELA project) 2026